



Social Media Analytics

Automated scraping, trend-detection and sentiment/demographic intelligence pipeline for X, Telegram, Instagram & more

PROBLEM STATEMENT ID

SIH26152

THEME

Miscellaneous

CATEGORY

Software

TEAM NAME

Status 200

TEAM LEADER

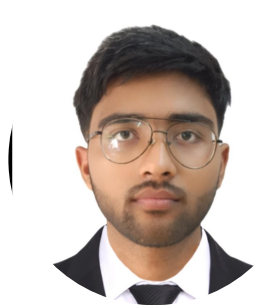
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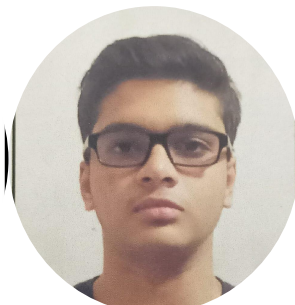


Team Details



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Proposed Solution

1

The Problem

Social media analysis today is siloed, i.e., sentiment, demographics, trends and influence are studied separately, giving an incomplete picture of how communities actually think and move.

2

Our Solution

An AI Driven, multi platform pipeline that goes through X, Telegram, Instagram and Facebook, then runs sentiment inference, demographic profiling, trend detection and network analysis on the same time, plus an AI Chatbot dashboard for querying insights

3

What Makes It Different

Built for link analysis and influence network mapping, not just sentiment for marketing metrics. Unlike existing platforms, our solution focuses on understanding the relationship, connections, and influence patterns within social networks, making these insights accessible to everyone.



Novelty and Uniqueness

EXISTING SOLUTIONS

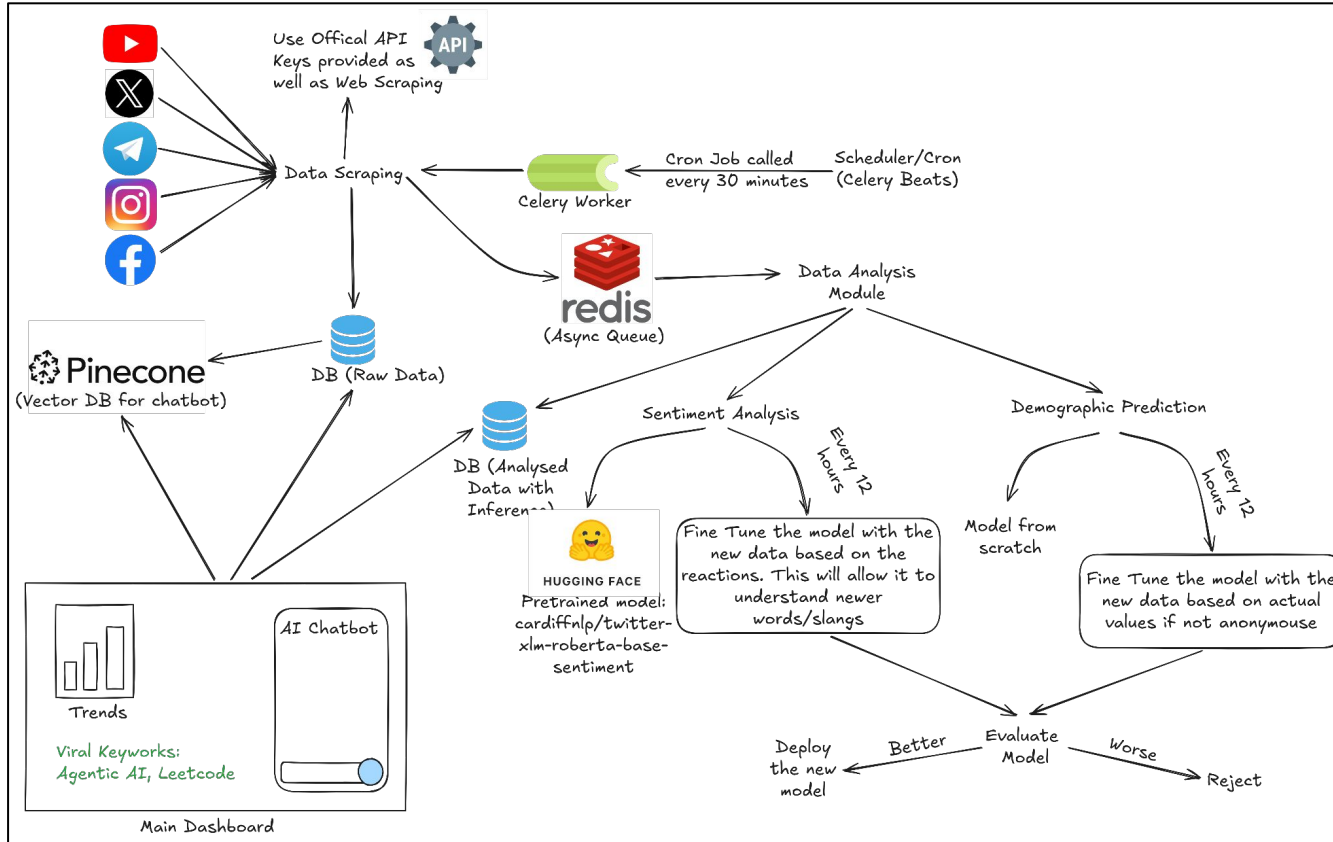
- [Brandwatch](#) / [Sprinklr](#) — strong sentiment & trend dashboards, but don't expose how their trend detection actually works.
- [Talkwalker](#) / [Meltwater](#) — good multi-platform coverage, but demographic inference is shallow and rarely explained.
- [Native platform analytics \(X/Meta\)](#) — accurate for a single platform only, with no unified raw-data store.

WHAT'S GENUINELY NEW

- A transparent viral-keyword algorithm (recent window → clean text → rank repeats) instead of a black-box trend score.
- A raw-information database anyone can query directly, plus a separate analysis database for derived insights.
- Judges/analysts see exactly which posts fed a trend, and can swap in pretrained or self-trained models per module.



Technical Approach



Backend



Frontend



AI/ML





Feasibility and Viability



Potential Challenges

- API rate limits & restricted/paid access on X and Meta platforms
- Scraping fallback (when APIs are unavailable) is fragile to layout/ToS changes
- Keeping the scraping and analysis queues near-real-time as post volume grows



Mitigation Strategy

- Prioritise official APIs (X API, Telegram Bot API), with 3rd-party APIs and scraping as fallback tiers
- Versioned scrapers with monitoring & alerts, so a broken scraper degrades gracefully, not silently
- Horizontally scalable Celery workers on both the scraping queue and the analysis queue



Impact and Benefits



Social Impact

Early warning for misinformation, public unrest, or community distress via real time sentiment analysis. It will also help organizations respond to the public concerns faster and more accurately. There might be risk of potential misuse of personal information for individual profiling, so an anonymised, aggregate only outputs and not individually targeted.



Economic Impact

Dependence on costly third party analytics vendors, sharpens ad spend by targeting real audience segments instead of assuming, also enables early product or campaign pivots by catching demand shifts or backlash before launch. Cuts potential brand damage costs through faster crisis response and opens new insight as a service revenue streams. It will also help in Job Creation in AI/Data roles needed to build and maintain the system.



Community Impact

Enables organisations to monitor and understand the needs, opinions and concerns of different communities through aggregate social media trends. Gives smaller or less heard communities a way for their concerns to be identified and considered in decision making. Encourages more responsive communication between organisations and the communities they serve. It can also help support diverse linguistic communities and give diverse language groups a voice.



Research and References

REFERENCES

- [*Big data analytics meets social media: A systematic review of techniques, open issues, and future directions*](#) - Sepideh Bazzaz Abkenar & Mostafa Haghi Kashani
- [*Social media analytics: a survey of techniques, tools and platforms*](#) - Bogdan Batrinca & Philip C. Treleaven
- [*Mining Social Media Data for Sentiment Analysis and Trend Prediction*](#) - SL Vijaya Durga & Anandhi R J
- Datasets Explored - [dair-ai/emotion](#) , [cardiffnlp/tweet_eval](#)



Solution Summary

Criterion	Summary
Problem	<i>Emerging trends and viral content spread faster than manual monitoring can track the sentiment or demographics behind them.</i>
Proposed Solution	<i>A scraping-and-queue pipeline (cron + Celery/Redis) that surfaces growing trends, viral keywords, sentiment and demographic predictions on a live dashboard.</i>
Novelty / USP	<i>A transparent viral-keyword algorithm plus swappable pretrained/on-the-go models for sentiment and demographics — not a black-box score.</i>
Feasibility	<i>Built on mature open-source tech (Django, Celery, Redis, PostgreSQL, HuggingFace); MVP scoped to raw scraping + dashboard first.</i>
Impact	<i>Analysts and researchers get faster, transparent insight into what is trending, why, and who it resonates with.</i>